

Problem Set 5

Costs and the Supply Curve

EC306: Intermediate Microeconomics

Instructions. Work through these in class. Show all working and interpret your numerical answers in words. Worked solutions follow at the end.

Question 1 — From technology to a cost function

A firm produces with $q = L^{1/2}K^{1/2}$, facing input prices w (labour) and r (capital).

- (a) Set up the cost-minimisation problem and use the tangency condition $MRTS = w/r$ together with the production constraint to derive the *conditional input demands* $L^*(w, r, q)$ and $K^*(w, r, q)$.
- (b) Show that the long-run cost function is $C(q) = 2q\sqrt{wr}$. Hence write down long-run MC and long-run average cost, and state what kind of returns to scale this technology has.
- (c) Verify Shephard's lemma for labour: show that $\partial C / \partial w = L^*(w, r, q)$, and explain in one sentence why this derivative recovers the cost-minimising input quantity.
- (d) The cost function is homogeneous of degree 1 in (w, r) . State what this means and give the economic reason it *must* hold for *any* cost-minimising firm, regardless of technology.

Question 2 — Short-run curves and the supply decision

In the short run a firm has total cost $C(q) = \frac{1}{3}q^3 - 4q^2 + 20q + 36$.

- (a) Identify fixed cost and variable cost, and derive $MC(q)$, $AVC(q)$, and $ATC(q)$.
- (b) Find the output at which AVC is minimised and the value of $\min AVC$. Confirm that MC cuts AVC at that point.
- (c) State the firm's shutdown price and write its short-run supply curve $q(p)$ explicitly, including the price region over which it supplies zero.
- (d) The fixed cost is 36. Explain why it appears in the firm's *exit* decision but *not* in its short-run shutdown decision.

Question 3 — A shutdown decision under stress

- (a) A competitive firm faces $p = 12$. Its cost function gives $MC(q) = 2q$, $AVC(q) = q$, and a fixed cost of 50. Find profit-maximising output, profit (or loss), and state with reasoning whether the firm produces in the short run.
- (b) Now suppose this is the firm's persistent situation in the long run. What is its long-run decision, and what is the lowest price at which it would remain in the industry?
- (c) **Real data.** Find one documented case (industry report, news, or filing, cited with an access date) of a firm or plant that kept operating at a loss during a downturn, *or* one that shut down. Explain the decision in this course's language: was price above or below the relevant average-cost threshold, and was the cost being escaped fixed or variable?

Question 4 — The long-run envelope and scale

- (a) Explain why the long-run average cost curve is the *lower envelope* of the short-run ATC curves, and why LRAC touches each SRATC at only one output (not at the SRATC minimum, except at one special point).
- (b) **Real data.** Identify a real industry with a clearly U-shaped or L-shaped LRAC and cite evidence for the shape (e.g. minimum efficient scale estimates).
- (c) Connect the shape you found to the likely number of firms the market can support, in 2–3 sentences.